

SIMON OKOMO ALOO

BSc, MSc, PhD

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Profile summary

I am an interdisciplinary food and biomedical scientist working at the intersection of food systems, nutrition, lipid metabolism, and cardiometabolic health. I hold a BSc (First Class Honours) in Food Science and Nutrition from Karatina University, Kenya, and a Master's and PhD in Food Science and Biotechnology from Kangwon National University (KNU), Republic of Korea. I am currently pursuing a second PhD in Biomedicine at Monash University-Baker Heart and Diabetes Institute, Australia. I am the Founder of the Alliance for Cardiometabolic Health and Research-Kenya (ACMHR-K), a non-profit advancing research, education, and advocacy for the prevention and management of cardiometabolic diseases. I am a recipient of the Government of Korea Scholarships for Undergraduate (GKS-U), Korea Scholarships for Graduate (GKS-G) studies, Brain Korea (BK) Fellowship, Bring Excellent Student to Kangwon National University (BEST-KNU) Scholarship, Monash Graduate Scholarship (MGS), and Baker Bright Spark Scholarship. I have also received Korean Ministry of Education Outstanding Academic Achievement Award, KNU Outstanding Research Achievement Awards for both Master's and PhD research, Korean Society of Food Science and Nutrition (KFN) Excellent Award, Korean Society of Food Science and Technology (KoSFoST) Excellent Award, and Global Youth Leader Exchange Program Best Speaker's Award.

EDUCATION

2025-Date: Doctor of Philosophy (Biomedical Science), Monash University, Australia (Ongoing)

Research: Tracing lipid metabolism to uncover lipid mediated cardiometabolic disorder mechanisms

2022-2024: Doctor of Philosophy (Food Science and Biotechnology), Kangwon National University, Korea

Thesis: Ameliorative effects of *Pediococcus acidilactici* KCTC 15831BP fermented hempseed on obesity and obesity-induced Alzheimer's disease: insights from microbiome and metabolomics.

- GPA: 4.32/4.5 (96%): Top 1% of the class
- Outstanding Research Achievement Award

2020-2022: Master of Science (Food Science and Biotechnology), Kangwon National University, Korea

- Thesis: *Anti-obesity effect of alfalfa sprout fermented with Pediococcus acidilactici in Caenorhabditis elegans model*. GPA: 4.49/4.5 (99%): Top 1% of the class

- Outstanding Research Achievement Award

2014-2018: Bachelor of Science (Food Science and Nutrition), Karatina University, Kenya

Thesis: *Effects of processing on the biochemical profile and proximate composition of tempeh made from lablab bean.*

- GPA: 3.82/4 (96%)
- First Class Honors: Top 1% of the class

SCIENTIFIC RESEARCH

- **Current Research:** Developing and applying a stable isotope lipid tracer approach on individualized cultured cells from human cohort studies to reveal genetic regulation of lipid metabolism. Findings will help in identifying genetic drivers of lipid dysregulation in cardiometabolic diseases.
- **Previous Research:** Developed probiotic-fermented plant-based foods with therapeutic potential against obesity and obesity-associated neurodegeneration, using in vitro cell lines, *C. elegans* and mouse as study models, and integrating microbiome and metabolomics analyses to elucidate underlying mechanisms and functional bioactivity
- **Publication history:** Total Publications -30 | Total first author articles - 20 | Co-author articles-8 | Total original articles -18 | Total review articles-10 | Book chapters - 2 | Q1 publications -24 | Q2 publications - 2 | Articles in top 1% journal-1; Articles in top 10% journals-8 | Articles in top 20% journals-12 | Google Scholar: 954 citations | h-index:16
- **Scientific reviewer:** Peer reviewer for manuscripts submitted to: Springer Nature, Elsevier, American Chemical Societies (ACS), and Springer Nature Journals
- **Selected Publications:** Five first-author publications in leading international journals:
 1. **Aloo, Simon Okomo**, Monicah Ndunda, and Silas Ambundo. "Industrial Trans Fat Elimination in African Food Systems: Reformulation Strategies, Regulatory Progress, Gaps, and Opportunities." *Trends in Food Science & Technology* (2026): 106045. DOI: <https://doi.org/10.1016/j.tifs.2026.106045>. **Impact Factor: 17.4; Rank: 1**
 2. **Aloo, Simon Okomo**, Godfrey Mwititi, Louise Wanjiku Ngugi, and Deog-Hwan Oh. "Uncovering the secrets of industrial hemp in food and nutrition: The trends, challenges, and new-age perspectives." *Critical Reviews in Food Science and Nutrition* 64, no. 15 (2024): 5093-5112. DOI: <https://doi.org/10.1080/10408398.2022.2149468> **Impact Factor: 10.6; Rank: 7**
 3. Aloo, Simon Okomo, SeonJu Park, and Deog-Hwan Oh. "Impacts of germination and lactic acid bacteria fermentation on anti-nutrients, bioactive compounds, and selected functional properties of industrial hempseed (*Cannabis sativa* L.)." *Food Chemistry* 428 (2023): 136722. DOI: [10.1016/j.foodchem.2023.136722](https://doi.org/10.1016/j.foodchem.2023.136722). **Impact Factor: 10.4; Rank: 8**

4. **Aloo, Simon Okomo**, SeonJu Park, Timilehin Martins Oyinloye, and Deog-Hwan Oh. "Rheological properties, biochemical changes, and potential health benefits of dehulled and defatted industrial hempseeds after fermentation." *Food Chemistry* 439 (2024). Access: [138086. 10.1016/j.foodchem.2023.138086](https://doi.org/10.1016/j.foodchem.2023.138086). **Impact Factor: 10.4; Rank: 8**
5. **Aloo, Simon Okomo**, Kaliyan Barathikannan, and Deog-Hwan Oh. "Polyphenol-rich fermented hempseed ethanol extracts improve obesity, oxidative stress, and neural health in high-glucose diet-induced *Caenorhabditis elegans*." *Food chemistry: X* 21 (2024): 101233. Access: <https://doi.org/10.1016/j.fochx.2024.101233>. **Impact Factor: 9.2; Rank: 10**

INTERNATIONAL SCHOLARSHIPS

2026-date: Baker Bright Spark Top-Up Scholarship, Baker Heart and Diabetes Institute

2025-date: Monash Graduate Scholarship (MGS) award, Monash University

2025: Australian Government Research Training Program (AGRTP) Scholarship, University of Wollongong

2022-2024: Bring Excellent Students to Kangwon National University (BEST-KNU) scholarship

2022-2024: Brain Korea 21 (BK21) scholarship, Kangwon National University

2019-2022: Korean Government Graduate Scholarship (KGSP), Kangwon National University

2018: Global Korea Scholarship for Undergraduate students, Asia-Pacific Africa Co-operation Academy (apaca).

SELECTED OVERSEAS OUTSTANDING AWARDS

2025 – Outstanding PhD Student Award, Kangwon National University, South Korea

Recognised as the overall Outstanding PhD Student for the 2025 academic year; awarded 28 February 2025.

2024 - Excellence Award, Korean Society of Food Science and Nutrition (KFN)

Awarded for an outstanding research presentation in the Graduate Oral Presentation session at the 2024 KFN Conference.

2022 - Outstanding Master's Student Award, Kangwon National University, South Korea

Recognised as the top-performing Master's student for the 2022 academic year; awarded 28 February 2022.

2024 - Excellence Award, The Korean Society of Food Science and Technology (KoSFoST),

Awarded for an outstanding research presentation in the Graduate Oral Presentation session at the 2022 KoSFoST International Symposium and Annual Meeting.

2021 – National Outstanding Academic Achievement Award, Ministry of Education, Republic of Korea

Recognised as the top Master's candidate among Korean Government Scholarship (GKS) scholars; presented on 22nd December 2021 by the Deputy Prime Minister of the Republic of Korea.

2020 - Excellence Award, Korea Youth Work Agency, Republic of Korea

Awarded third place in an essay competition organised by the Korea Youth Work Agency; presented by the Agency's Chief Executive Officer.

2020 - Best Speaker Award, Global Youth Leader Exchange Program

Recognised for outstanding public speaking in a TED-style speech competition on "COVID-19 and Life Changes."

2018 - Excellence Award, Asia-Pacific African Academy (APACA)

Recognised as an outstanding participant in the 2018 Training Program for Young Upcoming African Agricultural Experts.

INTERNATION CONFERENCE: ORAL PRESENTATIONS

Aloo, Simon Okomo (2024). *Pediococcus acidilactici*-fermented hempseed alleviates obesity and improves Alzheimer's disease-like symptoms via oxido-inflammatory pathways [Oral presentation]. Korean Food Science and Nutrition (KFN) Society 2024 International Annual Meeting, **Jeju, South Korea.**

Aloo, Simon Okomo (2023). *Protective effect of fermented hempseed extracts against obesity and obesity-induced oxidative stress and neurodegenerative disorders* [Oral presentation]. 16th International Scientific Conference on Probiotics, Prebiotics, Gut Microbiota and Health, **Bratislava, Slovakia.**

Aloo, Simon Okomo (2023). *Combination of elicited germination and fermentation improves metabolite profile and anti-obesity properties of alfalfa seed* [Oral presentation]. Korean Society of Food Science and Technology (KoSFoST) International Symposium and Annual Meeting, **Busan, Korea.**

TRAINING

2023: *Kangwon National University*

Topic: Ethical guidelines for the use of experimental animals and related skills.

2021: *Netmission Academy.Asia, Hong Kong*

Topic: Internet governance and human rights online

2020: *International Youth Leader Exchange Program*

Topic: World economy post-COVID-19 era.

2018: *Asia Pacific-African Cooperation Academy*

Topic: Emerging technologies in agriculture.

PROFESSIONAL EXPERIENCE AND LEADERSHIP ENGAGEMENTS

November 2025–Present: Founder of Alliance for [ACMHR-K | Alliance for Cardio-Metabolic Health and Research – Kenya](#), a non-profit organisation advancing cardiometabolic health Africa through research, evidence translation, professional education, mentorship, policy engagement and multi-sector collaboration. The organisation provides a platform connecting scientific research with practical public-health and food-policy action

Key leadership and impact so far include:

- Led ACMHR-K's engagement with national stakeholders on the review and strengthening of Kenya's industrial trans-fat regulations. This work contributed to the evidence base for regulatory action and resulted in a publication in Trends in Food Science & Technology (2026), ranked #1 in Food Science & Technology (Impact Factor-17.4). Access: [ACMHR-K | Alliance for Cardio-Metabolic Health and Research – Kenya](#)
- Led an establishment of a CPD-accredited continuing medical education (CME) programme with Kitui County Referral Hospital addressing the prevention of cardiometabolic disease. Access: [ACMHR-K | Alliance for Cardio-Metabolic Health and Research – Kenya](#)
- Research capacity and mentorship. Access: Established the ACMHR-K Ambassador Training Program, providing structured training and mentorship to emerging health and research professionals. Access: [ACMHR-K | Alliance for Cardio-Metabolic Health and Research – Kenya](#)
- Led the Empowering Kenyan Women in Research, Career and Leadership Workshop, connecting early-career researchers with established scientists and mentors from Australia, the United States, Africa and Europe. Access: [ACMHR-K | Alliance for Cardio-Metabolic Health and Research – Kenya](#)
- International collaboration: Developed an international network of researchers, health professionals and organisations across Australia, Kenya, South Africa, Lithuania and the United States, creating opportunities for research collaboration, mentorship and knowledge exchange.
- Australian-Kenyan research and health links: Initiated collaborations between ACMHR-K and Australian organisations to strengthen community engagement, professional development and evidence-based health initiatives in Kenya.
- Kenyan first Food Lipidome Database (KFLD): Developing ACMHR-K's research programmes to generate locally relevant evidence on dietary lipids, food composition, cardiometabolic risk and healthier food systems, with the aim of supporting evidence-informed policy in Kenya and the wider African region: [ACMHR-K | Alliance for Cardio-Metabolic Health and Research – Kenya](#)

March 2026–Date: Monash University, Science, Technology, and Medicine (STM) Student mentor

- Provide peer mentorship to honor students by offering guidance on the research experience, fostering a supportive and inclusive environment, and helping mentees navigate university resources.

2025-Date: Chairperson, Alliance for Cardiometabolic Health & Research–Kenya (ACMHR-K).

- Provide leadership for a national non-profit focused on cardiometabolic disease prevention through interdisciplinary research, training, and policy advocacy (<https://acmhr-k.org/>).

2022-2024: Board Clerk & Campus Ministry Coordinator, Harvest Time Intl. Church, Korea.

- Supported needy student through campus ministry fellowship at both Hallym University and Kangwon National University campuses

2020-2021: Kangwon National University-Team Leader, International University Promotion Committee

- Led an international team in promoting Kangwon National University to prospective students and global partners, supporting international recruitment, visibility, and institutional outreach.

2020-2021: African Youth Entrepreneurship Academy-Mentor

- Mentored and trained young entrepreneurs and Agri-professionals across Africa, building capacity in agribusiness innovation, value addition, and sustainable agricultural entrepreneurship

2017: Food Science Intern - Kenya Bureau of Standards (KEBS), Kenya

- Assisted in food quality analysis and product standardization activities.
- Conducted laboratory testing for physicochemical properties of foods.
- Supported inspection and certification processes for locally manufactured food products.

2016: Clinical Nutrition Intern - Homa-Bay Referral Hospital, Kenya

- Assessed patients' nutritional status and developed individualized diet plans.
- Assisted in management of diet-related conditions (diabetes, hypertension, malnutrition).
- Participated in hospital feeding program evaluation and dietary monitoring.

EXPERTISE

Lipidomics and metabolomics | Stable-isotope lipid tracing | Cell culture | Mouse models of metabolic disease | *Caenorhabditis elegans* models | Gut microbiome | Nutritional biochemistry | Functional food research | Health policy.

REFEREES.